

Project Description - Dynamic Programming

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Who are the members of your research team? Jacob Asp Nissen, fcl290, and Albert Luis Føns Olsen, zmr385.

What is the (working) title of your paper?

What is the research question?

We are trying to replicate the findings of Della-Vigna et al (2017) about reference dependent job search in Hungary. If possible we will investigate whether a model with heterogenous agents fits the data better.

References to relevant literature that you are inspired by - best to point out one “key reference”.

Della-Vigna et al (2017). "Reference-Dependent Job Search: Evidence from Hungary.

Job Search Models

The pattern of exit hazards after joblessness is initially high, then decreases throughout the benefit period and right before benefit expiration it surges, to decline gradually afterwards again.

The standard Model as of Mortensen (1986) and Van den berg (1990) predicts increasing exit hazard until benefit exhaustion and thereafter a constant exit rate. This obviously doesn't fit the true data, so different alterations have been made.

1. Unobserved heterogeneity - The productive workers get a job first and the least productive workers remain without. Thus, the hazard rate falls as it is more difficult for less productive workers to find employment.

DellaVigna et al (2017) proposes a model that fits data without unobserved heterogeneity. Instead they use reference-dependence. So people are loss averse for their income below a reference-point. That reference-point is set to their recent earnings. After some time however they adjust to their lower earnings and their 'reference' is set similarly lower. Thus, explaining a fall in search effort and lower hazard rates over time.

Concepts: Hand-to-mouth consumption (initial assumption): meaning no savings. Everything earned is spent during the same period.

The model you have in mind as a starting point for your analysis. Dellavigna et al (2017) is working with two different models. The standard model, which is given as follows.

The observed agents is without a job and chooses his search effort in order to maximize utility. There is increasing costs associated with search effort given by the convex, twice differentiable and time separable function, $c(s_t)$, where search effort is given by $s_t \in [0; 1]$ and s_t equals the probability of having a job in

period $t + 1$. We consider a hand-to-mouth model where consumption equals wages in the same period. $c_t = y_t$. Dellavigna et al (2017) are also working with a model that include assets. The agents gets utility from consumption, $u(c_t) = v(c_t)$

The reference dependent model This model has the same properties as the standard model, but includes reference-dependence utility, such that utility comes from consumption as well as some gain-loss utility derived from the difference between consumption today and the reference consumption.

$$u(c_t) = \begin{cases} v(c_t) + \eta[v(c_t) - v(r_t)] & \text{if } c_t \geq r_t \\ v(c_t) + \eta\lambda[v(c_t) - v(r_t)] & \text{if } c_t \leq r_t \end{cases} \quad (1)$$

Where the reference is given as an arithmetic average of the income (and thus consumption in the hand-to-mouth model) in the N latest periods.

$$r_t = \frac{1}{N} \sum_{k=t-N}^{t-1} y_k$$

Rather than assuming hand-to-mouth consumption, we allow individuals to accumulate or run down assets in order to "adjust/smooth" consumption in the current period. We denote the budget constraint that the individual faces is $\frac{A_{t+1}}{1+R} = A_t + y_t - c_t$. Introducing assets allows for individuals to have different preferences towards search intensity as entering unemployment with more assets would allow for greater patience and thus increase the marginal cost of jobsearch. With Assets introduced we can write the value functions that we seek to maximize. For the unemployed the value function only depends on current assets and can be expressed as:

$$V_t^U(A_t) = \max_{s_t \in [0,1]; A_{t+1}} u(c_t|r_t) - c(s_t) + \delta[s_t V_{t+1|t+1}^E(A_{t+1}) + (1 - s_t)V_{t+1}^U(A_{t+1})] \quad (2)$$

subject to:

$$c_t = A_t + y_t - \frac{A_{t+1}}{1+R}$$

We can express the value function of an individual that is employed in period t and found the job in period j , where t and j combined determines reference-point, as:

$$V_{t|j}^E(A_t) = \max_{c_t > 0} u(c_t|r_t) + \delta V_{t+1|j}^E(A_{t+1})$$

Using the value function for unemployment we write the FOC for optimal level of search

$$c'(s_t^*(A_{t+1})) = \delta[V_{t+1}^E(A_{t+1}) - V_{t+1}^U(A_{t+1})]$$

1 Solving the different models

1.1 Hand-to-Mouth

1.2 Finding the optimal search effort

We use equation (2), the value function for the unemployed to solve for the optimal search effort by optimizing around the search cost function. / by taken the derivative with regards to s_t .

$$V_t^U(A_t) = \max_{s_t \in [0,1]; A_{t+1}} u(c_t|r_t) - c(s_t) + \delta[s_t V_{t+1|t+1}^E(A_{t+1}) + (1 - s_t)V_{t+1}^U(A_{t+1})] \quad (3)$$

The search cost function is given as $c_s = \frac{ks^{1+\gamma}}{1+\gamma}$. To be used later.

$$\begin{aligned} \text{FOC: } \frac{\partial V_t^U(A_t)}{\partial s_t} &= 0 \\ \Leftrightarrow -\frac{\partial c(s_t)}{\partial s_t} + \delta[V_{t+1|t+1}^E(A_{t+1}) - V_{t+1}^U(A_{t+1})] &= 0 \\ \Leftrightarrow c'(s_t^*(A_{t+1})) &= \delta[V_{t+1|t+1}^E(A_{t+1}) - V_{t+1}^U(A_{t+1})] \end{aligned}$$

So this is actually the marginal increase in search cost with the optimal search effort. This is not a function showing optimal search effort.

To find that we need to solve for s^* by inverting the function.

$$s_t^* = \mathcal{C}(\delta[V_{t+1|t+1}^E(A_{t+1}) - V_{t+1}^U(A_{t+1})]) \quad (4)$$

$$\text{where } \mathcal{C}(\cdot) = c'^{-1}(\cdot) \quad (5)$$

As $c_s = \frac{ks^{1+\gamma}}{1+\gamma}$ then $c'_s = ks^\gamma$ and the inverse function is:

$$c'(s) = ks^\gamma \quad (6)$$

$$\Leftrightarrow s = \left(\frac{c'}{k}\right)^{1/\gamma} = c'^{-1}(\cdot) = \mathcal{C}(\cdot) \quad (7)$$

So the optimal search effort becomes a function of the asset value in time $t+1$. Where the search effort decreases as patience shrinks, but grows as the net present value of having a job in the future rises compared to being unemployed.

1.3 Model with assets

We follow the recipe of solving the model proposed by DellaVigna et al (2017). First, we find the steady-state value of the employed state. Then we use backwards induction to find all the values of being employed, $V_{t|t}^E(A_t)$, given all the different asset values. After that we find the value of being unemployed with different asset values. We solve for the optimal search effort path and consumption path for the unemployed too.

We start by finding the steady-state value of being employed. The value of function of employment when

$c_t = r_t$ is given by:

$$V_t^E(A_t) = \max u(c_t) + \delta V^E(A_{t+1}) \quad (8)$$

This means that the value of consumption doesn't change due to different preferences and thus we can find a steady-state level. Thus the bellman equation here can be solved as a standard consumption-saving problem with infinite horizon and constant income. We find the optimal level of consumption.

Remember the full sketched out value function, where we substitute consumption for the budget constraint. Remember we are in the employed state and wages are fixed know, so there is no need for a time constraint there.

$$u(c_t) + \delta V^E(A_{t+1}) = \ln(A_t + w - \frac{A_{t+1}}{1+R}) + \delta V^E(A_{t+1}) \quad (9)$$

So now we take the first order condition

$$\begin{aligned} \text{FOC: } \frac{\partial u(c_t)}{\partial A_{t+1}} + \delta \frac{\partial V_{t+1}^E(A_{t+1})}{\partial (A_{t+1})} &= 0 \\ \Leftrightarrow \\ \frac{1}{A_t + w - \frac{A_{t+1}}{1+R}} \cdot \left(-\frac{1}{1+R}\right) + \delta \frac{1}{A_{t+1} + w - \frac{A_{t+2}}{1+R}} &= 0 \\ \Leftrightarrow \\ \frac{1}{A_t + w - \frac{A_{t+1}}{1+R}} &= \delta(1+R) \frac{1}{A_{t+1} + w - \frac{A_{t+2}}{1+R}} \end{aligned}$$

Which can also be written as the derivative of the utility function for c_t and c_{t+1} . Thus we write.

$$u'_c(c_t) = \delta(1+R)u'_c(c_{t+1}) \quad (10)$$

If we impose the standard assumption that $\delta(1+R) = 1$ then $u'_c(c_t) = u'_c(c_{t+1})$ and thus $c_t = c_{t+1} = c^*$. If consumption don't differ then the budget constraint imposes that the asset levels are constant too.

If neither differs then we can find the steady-state level from the budget constraint:

$$\begin{aligned} c_t &= A_t + w - \frac{A_{t+1}}{1+R} = c_{t+1} = c^* \\ \Leftrightarrow \\ c^* &= A + w - \frac{A}{1+R} = w + \frac{R}{1+R} \cdot A \end{aligned}$$

So, the steady state consumption is given by the constant wages and plus the annual return of assets. This means that for a given asset level then we can find the value of being employed as a perpetuity, geometrical row.

$$V^E = u(c^*) + \delta V^E = u(c^*) + \delta u(c^*) + \delta^2 V^E \dots \quad (11)$$

$$= \frac{u(c^*)}{1 - \delta} \quad (12)$$

Or if you want to look at it for different asset values it becomes $V^E(A) = \frac{u(c^*(A))}{1 - \delta}$ We solve that for every asset level.

We plug that value into the value of unemployment.

$$V_t^U(A_t) = \max_{s_t \in [0,1]; A_{t+1}} u(c_t | r_t) - c(s_t) + \delta \left[s_t \frac{u(c^*(A_{t+1}))}{1 - \delta} + (1 - s_t) V_{t+1}^U(A_{t+1}) \right] \quad (13)$$

1.4 Solving the model with assets

We have already found the steady state value of employment to be:

$$\text{Steady State Value: } \frac{w + \frac{R}{1+R} A_T}{1 - \delta}$$

Where T is set as the time period from where assets are constant at the steady state level. From there on we can solve the value of A_{T-1} by solving for the two period consumption-saving model, where the second value of consumption is the value of the discounted period of all future consumption. $C_t = \frac{w + \frac{R}{1+R} A_T}{1 - \delta}$.

$$\max V_t = \max u(c_{T-1}) + \delta \max u(c_T)$$

by substituting in C_T and the budget constraint for c_{T-1}

$$\max V_t = \max u \left(w + A_{T-1} - \frac{A_T}{1+R} \right) + \delta u \left(\frac{w + \frac{R}{1+R} A_T}{1-\delta} \right)$$

Taken the first order condition wrt. A_T , as $u = \ln(\cdot)$:

$$-\frac{1}{(1+R)(w + A_{T-1}) - A_T} + \delta \frac{1-\delta}{w + \frac{R}{1+R} A_T} \cdot \frac{\frac{R}{1+R}}{1-\delta} = 0$$

$$\Leftrightarrow$$

$$\frac{1}{(1+R)(w + A_{T-1}) - A_T} = \delta \frac{\frac{R}{1+R}}{w + \frac{R}{1+R} A_T}$$

$\Leftrightarrow$ Vi ganger med $(1+R)$ i tæller og nævner

$$\frac{1}{(1+R)(w + A_{T-1}) - A_T} = \frac{\delta R}{(1+R)w + RA_T}$$

$$\Leftrightarrow$$

$$(1+R)w + RA_T = \delta R(1+R)(w + A_{T-1}) - \delta RA_T$$

$$\Leftrightarrow$$

$$(1+R)w + (1+\delta)RA_T = \delta R(1+R)(w + A_{T-1})$$

$$\Leftrightarrow$$

$$(1+R)w - \delta R(1+R)w + (1+\delta)RA_T = \delta R(1+R)A_{T-1}$$

$$\Leftrightarrow$$

$$A_{T-1} = \frac{(1+R)w - \delta R(1+R)w + (1+\delta)RA_T}{\delta R(1+R)}$$

$$\Leftrightarrow$$

$$A_{T-1} = \frac{1}{\delta R}(1-w) + \frac{1+\delta}{\delta(1+R)}A_T$$

So now we have the optimal level of A_{T-1} , ie. the level of assets you choose to hold in the second last period given the asset level in the last period. As A_{T-1} holds the budget constraint we can see this as the correct solution to the two period problem.

So, now this was conceptually the difficult problem. Now when we know the solution to the optimal level of assets in the second last period we can always use that to find the asset level in the third last period by optimizing the normal Value Function of employment given A_{T-1} . Do this as the first thing tomorrow!

2 Future outline

The solution methods and numerical techniques you consider applying

In order to solve the model numerically we backwa

The estimation mention methods you consider on implementing - Which one do they use?

We estimate our parameter by employing a Generalized Method of Moments framework, using a weighted least squares estimator to choose the parameter that minimizes the distance between the predicted/expected moment and our observed moment.

The data you rely on (if empirical paper)

We are hoping to get access to the same data that is used in the reference article. We have send an email to the data providers. It is unlikely we receive the data. Instead we expect to be able to simulate some of the data from some of the aggregated data that we have access to.

Potential counterfactual simulations of interest

A progressive plan of action / plan of work (start simple).

Something we could do to extend the model:

- Change the reference so that you weigh the latest points more
- Heterogenous agents (how to really do that?)

2.1 Things to check for

1. How vulnerable is the model to changes in the reference point period? Are these vulnerabilities feasible?
2. How vulnerable is the model towards changes in the discount factor?
3. How vulnerable is the model towards changes in the relative wage between unemployment and employment?
4. Different values for η and λ too